

Biology

Middletown High School | Grade 12, Fall Semester 2026

Teacher: Ms. R. Castellanos
Room: 117 Lab
Email: rcastellanos@middletownhigh.edu

Course Description

A survey of life from the cell to the ecosystem, with microscope work, dissection and a field study. Students practice reading data and explaining what a result does and does not show.

Course Units

Weeks 1 to 3	Unit 1: Cell Structure and Function Prokaryotic and eukaryotic cells. Nucleus, mitochondria for respiration, ribosomes for protein synthesis and chloroplasts for photosynthesis in plants. Diffusion, osmosis and active transport.
Weeks 4 to 7	Unit 2: Genetics and Inheritance DNA as a double helix with base pairs A to T and C to G. Mitosis makes identical body cells, meiosis makes gametes. Punnett squares predict offspring ratios for dominant and recessive alleles.
Weeks 8 to 10	Unit 3: Evolution and Natural Selection Variation, selection pressure, speciation and the evidence from fossils, anatomy and DNA.
Weeks 11 to 13	Unit 4: Human Body Systems Circulatory, respiratory, digestive and nervous systems, and how they maintain homeostasis.
Weeks 14 to 16	Unit 5: Ecology Energy flow through food webs, nutrient cycles, population growth and human impact.

Key Dates This Semester

Sep 24	Microscope Lab
Oct 19	Mid-Term Exam, covers Units 1 to 2
Nov 12	Ecology Field Report due
Dec 15	Final Exam

School wide dates

Sep 8	First day of classes
Oct 19 to Oct 23	Mid-term examination week
Nov 6	Parent teacher conferences
Nov 25 to Nov 27	Thanksgiving break, no classes
Dec 14 to Dec 18	Final examination week

Materials

Lab notebook, colored pencils for diagrams, scientific calculator.

Grading Policy

20 percent	Laboratory and field work
15 percent	Homework
15 percent	Quizzes
30 percent	Unit tests
20 percent	Final exam

A = 90 to 100 percent, B = 80 to 89, C = 70 to 79, D = 60 to 69, F = 59 and below.

Late and Make-Up Work

Work handed in late without an excused absence receives a maximum of 50 percent credit. With an excused absence, work is due within one week of returning. It is the student's responsibility to ask what was missed. Make-up tests are held on Thursdays after school and must be booked at least one day ahead.

Academic Honesty

All work submitted must be your own. Copying another student's work, or submitting text or code generated by someone or something else as your own, is treated as academic dishonesty and results in a zero for that assignment and a call home.

Extra Help

Extra help is available Tuesdays and Thursdays, 3:15 to 4:00 pm, in the room listed above.